

Micro-Lending & Collection Tracking Application

This document outlines the core concept, architectural workflow, and functional process flow for a comprehensive micro-lending and daily collection management system. Designed to streamline operations for local financiers, the platform supports highly configurable loan packages, automated penalty calculations, and strict role-based access controls to ensure data integrity across various geographic collection routes, such as the Erode and Chithode lines.

1. System Roles & Access Control

The system utilizes a dual-tier authentication architecture to maintain financial security while allowing flexible field operations.

Role	Responsibilities & Permissions
Data Entry / Field Agent	Can log daily/weekly collections from customers. Constraint: Once a payment is submitted, the agent cannot edit, modify, or delete the entry.
Admin / Manager	Full access to system configurations, penalty overrides, package creation, and correction of erroneous entries submitted by field agents.

2. End-to-End Process Flow

Phase 1: Customer Onboarding & KYC

- Profile Creation:** A new customer profile is initiated with standard demographic details (Name, Phone Number, Address).
- ID Generation:** The system assigns a unique alphanumeric tracking code. The prefix can denote the location and frequency (e.g., ER-DLY-001 for Erode Daily).
- Document Vault:** The agent uploads mandatory KYC documents (Aadhar) and registers up to five security cheques (logging Bank Name and Cheque Number).

Phase 2: Loan Allocation & Disbursement

1. **Package Selection:** The Admin selects the collection frequency (Daily, Weekly, or Monthly) and sets the tenure (e.g., 100 days, 10 weeks).
2. **Principal Deduction:** The system calculates the internal deduction (interest or processing fee). For example, on a ₹30,000 loan, a ₹300 deduction is logged, and the net cash disbursed is recorded via a debit voucher mapped to the customer's ID.
3. **Schedule Generation:** The system automatically projects the payment schedule, populating exact calendar dates for the entire term.

Phase 3: Collection Tracking

- **Active Logging:** As collections happen, agents input the received amount against the scheduled date. The outstanding balance automatically recalculates.
- **Missed Payments:** If a payment is not received on a scheduled date, the system leaves that date slot **empty** to visibly highlight the default, rather than pushing the date forward.
- **Midnight Cutoff:** Daily ledgers lock and rollover at midnight (12:00 AM), finalizing the day's collection status.

Phase 4: Penalty Calculation Engine

- **Automated Accrual:** For every empty slot (missed payment), the system automatically accrues a predefined penalty (e.g., ₹10/day).
- **Accumulation & Visibility:** Penalties are tracked in a separate ledger column or view to prevent them from superficially inflating the core principal outstanding balance, ensuring clean accounting.
- **Settlement/Waiver:** At the end of the term, the Admin can either enforce the total accumulated penalty, negotiate a partial settlement, or "neutralize" (waive) the penalty to close the account cleanly.

Phase 5: Loan Closure & Renewal

- **Settlement:** Once the principal and any enforced penalties reach zero, the loan is marked as "Closed."
- **Security Release:** The system prompts the return or disposal of the registered security cheques to the customer.
- **Renewal:** If the customer requests a new loan, the existing User ID is maintained, and a new active ledger schedule is generated. The previous loan data is archived into the customer's historical view.

3. Dashboard & Analytics

The centralized admin dashboard provides real-time oversight of the financial health of the lending operation.

- **Daily Snapshot:** Displays total expected collections vs. actual received amount for the current day.
- **Defaulter List:** A real-time tracker showing customers who have missed their recent payments, quantifying the exact number of missed days and accumulated penalty.
- **Route Grouping:** The ability to filter the dashboard by specific geographic lines or groups (e.g., Erode, Chithode) to monitor the performance of specific field agents or territories.